



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 wide notification number F-9-12/2001-U-3 of Government of India)

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Computer Networks/0705210503

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Internet Control Message Protocol (ICMP)

- The protocol is at the network layer.
- ICMP is used for reporting errors and management queries.
- Since IP does not have an inbuilt mechanism for sending error and control messages. It depends on the Internet Control Message Protocol(ICMP) to provide error control.
- It is a supporting protocol and is used by network devices like routers for sending error messages and operations information.

Example: The requested service is not available, or a host or router could not be reached.

- Since the IP protocol lacks an error-reporting or error-correcting mechanism, information is communicated via a message.
- If a middleman reports the mistake, ICMP helps in notifying the sender about the issue.



Internet Control Message Protocol (ICMP)

- ICMP is used for error reporting if two devices connect over the internet and some error occurs, so the router sends an ICMP error message to the source informing about the error.
- Another important use of the ICMP protocol is to perform network diagnosis by making use of *traceroute* and the *ping* utility.

Traceroute: Traceroute utility is used to know the route between two devices connected over the internet. It routes the journey from one router to another, and a traceroute is performed to check network issues before data transfer.

Ping: Ping is a simple kind of traceroute known as the echo-request message, it is used to measure the time taken by data to reach the destination and return to the source, these replies are known as *echo-replies* messages.

How Does ICMP Work

- ICMP is the primary and important protocol of the IP suite, but ICMP isn't associated with any transport layer protocol (**TCP or UDP**) , as it doesn't need to establish a connection with the destination device before sending any message, as it is a connectionless protocol.
- The working of ICMP is just contrasting with TCP, as TCP is a connection-oriented protocol, whereas ICMP is a connectionless protocol.
- ICMP packets are transmitted in the form of datagrams that contain an IP header with ICMP data. An ICMP datagram is similar to a packet, which is an independent data entity.

ICMP Packet Format

Type(8 bit)	Code(8 bit)	CheckSum(16 bit)
Extended Header(32 bit)		
Data/Payload(Variable Length)		



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In the ICMP packet format, the first 32 bits of the packet contain three fields:

Type (8-bit): The initial 8-bit of the packet is for message type, which provides a brief description of the message so that the receiving network would know what kind of message it is receiving and how to respond to it.

Some common message types are as follows:

- Type 0 - Echo reply
- Type 3 - Destination unreachable
- Type 5 - Redirect Message
- Type 8 - Echo Request
- Type 11 - Time Exceeded
- Type 12 - Parameter problem



- The **Echo Request** is sent to a target device to ask, "Are you there?" or "How long does it take to communicate with you?"
- The **Echo Reply** is the response sent back by the target device saying, "Yes, I am here," along with relevant information like timestamps or sequence numbers.
- The **destination is unreachable** and is generated by the host or its inbound gateway to inform the client that the destination is unreachable for some reason.
- **Redirect requests** data packets are sent on an alternate route. The message informs a host to update its routing information (to send packets on an alternate route).
- **Parameter Problem:** Whenever packets come to the router then the calculated header checksum should be equal to the received header checksum then only the packet is accepted by the router.



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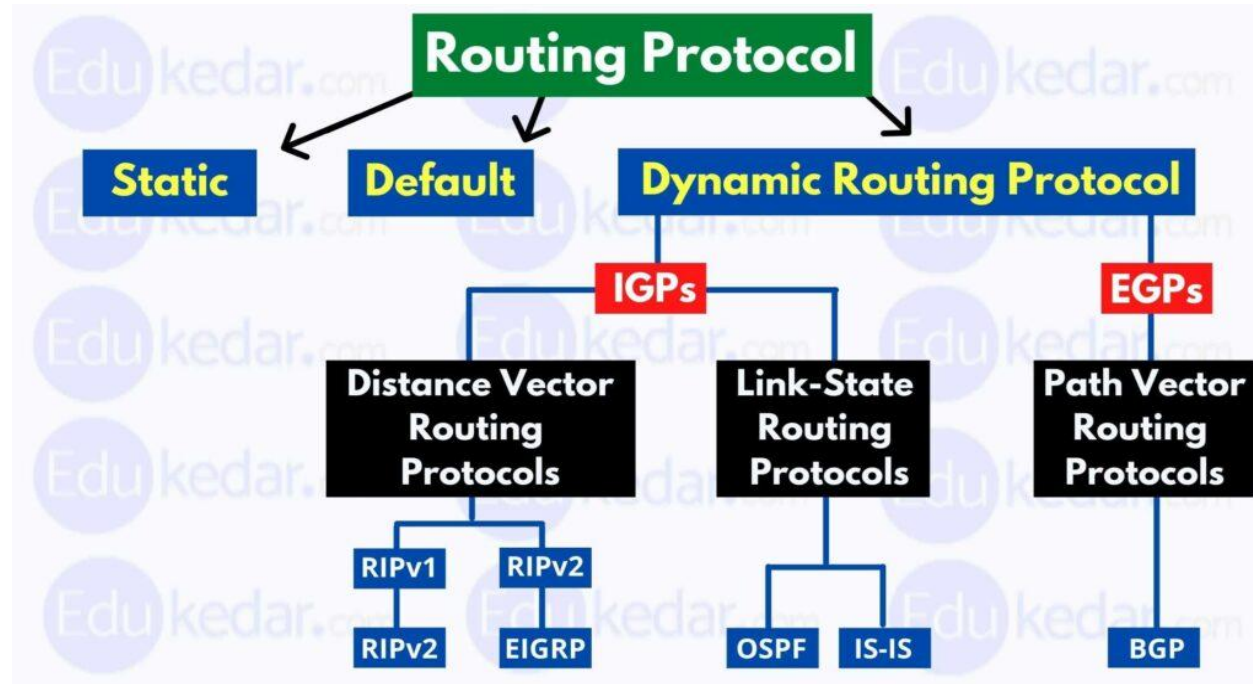


- **Code (8-bit):** Code is the next 8 bits of the ICMP packet format, this field carries some additional information about the error message and type.
- **Checksum (16-bit):** Last 16 bits are for the checksum field in the ICMP packet header. The checksum is used to check the number of bits of the complete message and enable the ICMP tool to ensure that complete data is delivered.
- The **next 32 bits** of the ICMP Header are Extended Header which has the work of pointing out the problem in IP Message.
- Byte locations are identified by the pointer which causes the problem message and receiving device looks here for pointing to the problem.
- The last part of the ICMP packet is **Data or Payload** of variable length. The bytes included in IPv4 are 576 bytes and in IPv6, 1280 bytes.



Routing Protocol

- Routing Protocols are simply a set of defined rules used by the routers to communicate between source & destination.
- These protocols do not move the information from the source to a destination, but they update the routing table that contains the information.





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Routing Protocol

- RIP- Routing Information Protocol
- OSPF- Open Shortest Path First
- IGRP- Interior gateway routing Protocol
- EIGRP- Enhanced Interior gateway routing Protocol
- IS-IS- Intermediate system to intermediate System
- BGP- Border Gateway Protocol



Routing Protocol

Routing Protocols can be divided in many ways. But in general, there are three categories that are as follows.

1. Static Routing Protocol
2. Dynamic Routing Protocol
3. Default Routing Protocol

Static Routing Protocol

When an administrator manually allocates the path from the source to the destination network, static routing protocols are employed. It improves the network's security.

- There is no load on the router's CPU.
- Between links, there is no unused bandwidth.
- Routes can only be added by the administrator.
- Because it is time-consuming, it is not a good choice for large networks.
- When a connection fails, the entire network goes down, which is impractical in small networks.



Dynamic Routing Protocol

- It enables routers to automatically add information from connected routers to their routing tables.
- When the topological structure of the network changes, these protocols send out topology updates.

Benefits:

- Even on larger networks, it's easier to set up.
- It will be able to choose an alternate route dynamically if a link fails.
- It assists you in load balancing over many links.

Disadvantages

- Because updates are exchanged among routers, bandwidth is used.
- Protocols increase the load on the router's CPU or RAM.